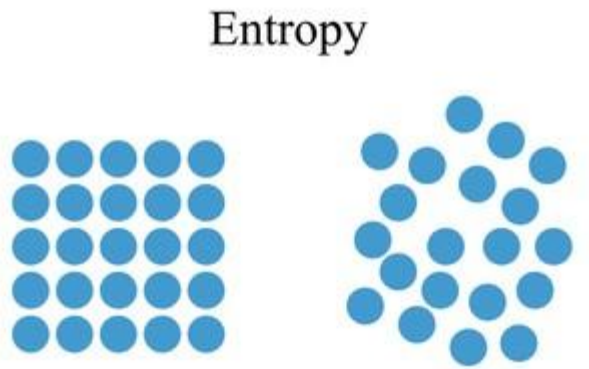
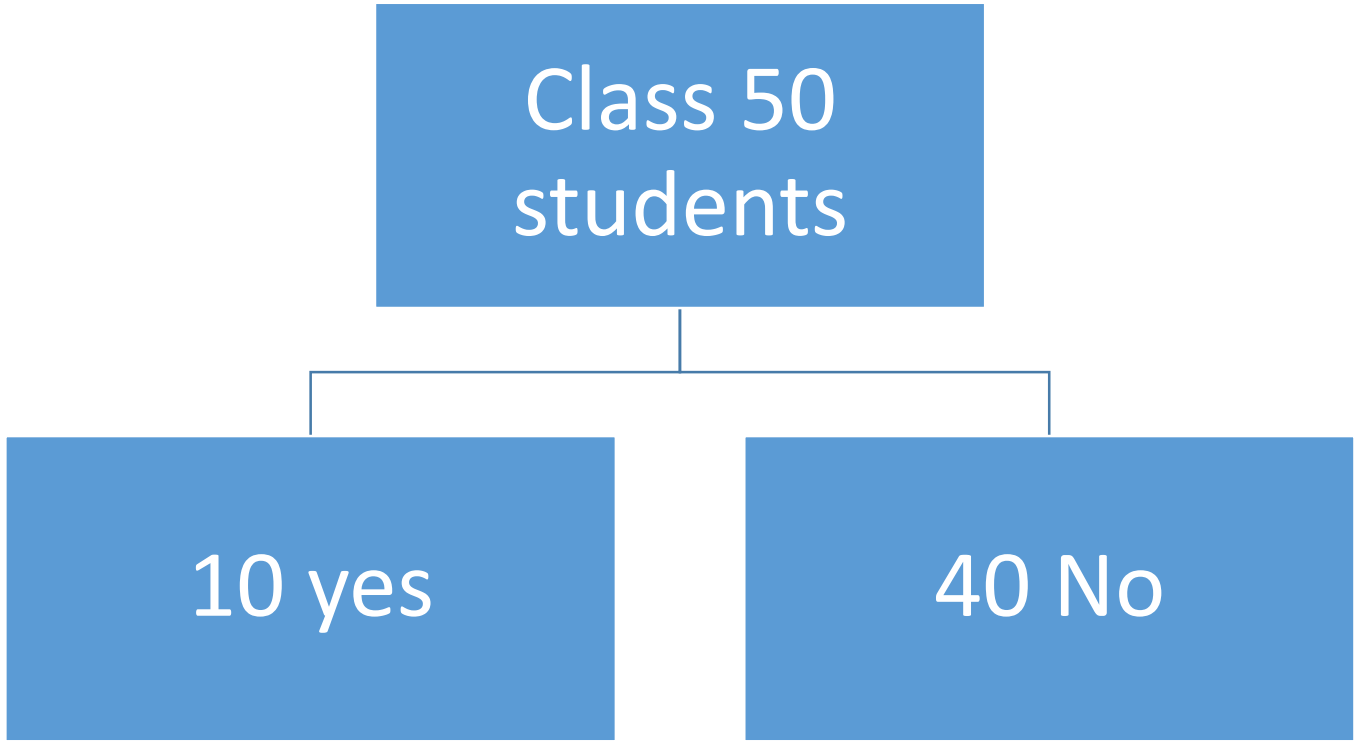
In Machine Learning, **entropy measures the amount of impurity, chaos, or uncertainty** in a dataset.
It tells you how mixed up your data classes are. A dataset with only one class has **zero entropy** (pure), while an equal mix of classes has **maximum entropy** (impure).




Class 50 students
10 yes
40 No

Entropy : **it defines Randomness and disorders**

$$Entroy = -\sum_{I=1}^{N} \boldsymbol{p_i \, log_2 \, p_i}$$ (9y , 5n)

Entropy of Root Node(9y , 5n)

$$Entroy = (-\mathrm{P}_y) * (\log_2(\mathrm{P}_y)) + (-\mathrm{P}_n) * (\log_2(\mathrm{P}_n))$$

$$= \left(-\left(\frac{9}{14}\right)\right) * \left(\log_2\left(\frac{9}{14}\right)\right) + \left(-\left(\frac{5}{14}\right)\right) * \left(\log_2\left(\frac{5}{14}\right)\right)$$

$$= (-0.64) * (\log_2(0.64) + (-0.35) * (\log_2(0.35))$$

$$= (-0.64) * (-0.64) + (-0.35) * (-1.49)$$

$$\boldsymbol{Entroy = 0.94}$$

Entropy Scale (For Binary Classification)

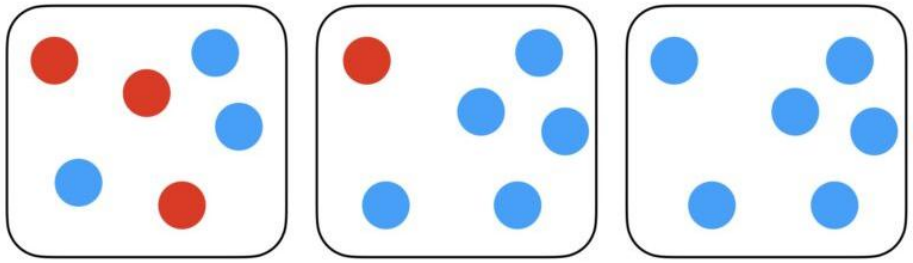
Entropy = 0 (Pure): All data points belong to a single class (e.g., 14y, 0n). The outcome is 100% predictable.
Entropy = 1 (Impure): Data points are split evenly between classes (e.g., 7.5y, 7.5n). Uncertainty is at its highest.
0 < Entropy < 1: A partial mixture (e.g., 9y, 5n gives 0.94, 2y, 12n gives 0.59).

Gini Impurity is a measure used in decision trees to determine how impure or mixed a dataset is. It represents the probability of incorrectly classifying a randomly chosen sample if it is labeled according to the class distribution in the dataset.

$$Gini = 1 - \sum_{i=1}^{n} (p_i\ )^2$$

(9y , 5n)

$$Gini = 1 - \left[\left(\frac{y}{Total}\right)^2 + \left(\frac{n}{Total}\right)^2 \right]$$



GI of Root Node(9y , 5n)

$$Gini = 1 - \left[\left(\frac{9}{14}\right)^2 + \left(\frac{5}{14}\right)^2 \right]$$

$$= 1 - [\ (0.64)^2 + (0.36)^2\ ]$$

$$= 1 - [\ 0.41 + 0.13\ ]$$

$$\boldsymbol{Gini\ = 0.46}$$